MANIPAL ACADEMY of HIGHER EDUCATION **Question Paper - Report**

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Question Paper

MANIPAL ACADEMY OF HIGHER EDUCATION

III Semester BTech Mid Term Examination - September 2024
COMPUTER ORGANIZATION and ARCHITECTURE [CSE 2121]

Marks: 30 **Duration: 90 mins.**

MCQs

Answer all the questions. Section Duration: 20 mins

1) The hardwired controller in Booth's multiplication algorithm has -------- inputs and ----------output pins. (0.5)

5, 10 5, 9 9, 5 10, 5

2) The disadvantage(s) of the hardwired approach is _____ (0.5)

It is less flexible It cannot be used for complex instructions It is costly Less flexible and cannot be used for complex instructions

3) Based on which of the following statements, the control signal is generated? (0.5)

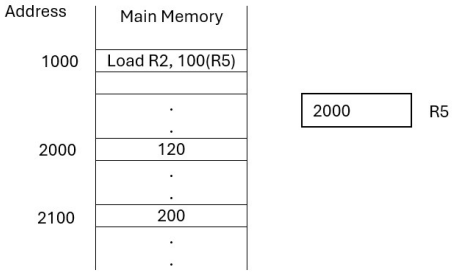
i. Contents of IR

ii. Contents of condition flags

iii. Program Counter

Only i and iii Only ii and iii Only i and ii i, ii and iii

4) Referring to the memory and register contents shown in the following figure, the content of register R2 after executing the instruction at address 1000 is _____ (0.5)

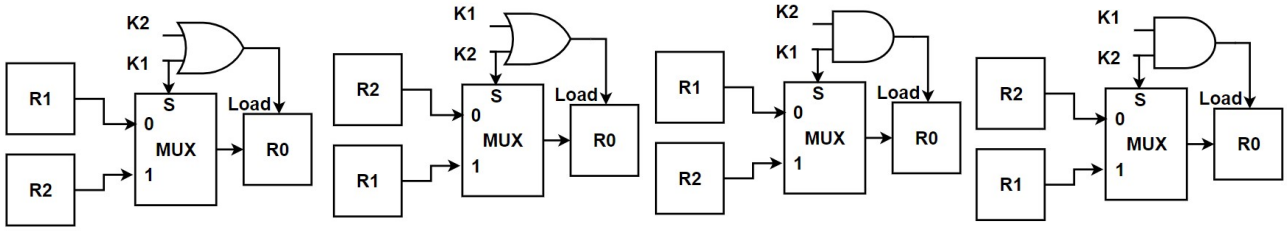


2000 120 2100 200

5) To perform M *(01111010), Booth's algorithm does ___ additions and ___ subtractions respectively. (0.5)

1 and 2 2 and 1 2 and 2 3 and 2

6) Correct hardwired implementation of the register transfer operation, K1: $R0_{\text{leftwards arrow}}R2$, K1'K2: $R0_{\text{leftwards arrow}}R1$ is ____ (0.5)



7) Which of the following correctly describes the trade-off between two-bus and three-bus architecture systems? (0.5)

Three-bus architectures are always more cost-efficient than two-bus architectures Two-bus systems provide faster data throughput but suffer from bus contention, while three-bus systems reduce contention but increase hardware complexity Two-bus architectures simplify the design by reducing the number of pathways, but this leads to higher bus contention than in one-bus systems All architectures provide similar performance under the same workload

8) Which of the following registers in the processor will have the instruction being executed? (0.5)

PC MBR MAR IR

9) A floating point 32-bit representation: 00000000101010111001101110111000 represents___ (0.5)

Underflow value Overflow value +infinity Floating point number within the range

10) With respect to the state diagram in hardwired control unit design, which of the following is true? (0.5)

The independent operations completed in clock cycle can be in one state Each operation must have independent state The independent operations completed in clock cycle cannot be in one state Conditional testing requires no state.

Descriptive

Answer all the questions.

11) With the help of flow chart, perform unsigned multiplication between two binary numbers where multiplicand = 101101 and multiplier = 10001. Show all the steps clearly. (4)

12) Show the detailed steps in multiplying the following two numbers using Booth's multiplication algorithm: M= +11 and Q= -16. (3)

13) Design a processing unit establishing the required control points for the Register Transfer Description (RTD) shown in Fig. P13. (3)

14) Design the block diagram of the controller and develop the corresponding state diagram for the RTD given in Fig. P13. (3)

15) Calculate the sum of two positive binary numbers 1110.1 and 10111.1 in IEEE 32-bit representation. Show all the steps clearly and represent the final answer in IEEE 32-bit format. (3)

16) Elaborate the single, two and three bus orientation for the CPU to perform $R2_{\text{leftwards arrow}}R0 + R1$. (3)

17) With a neat diagram, design and explain bi-directional data bus. (2)

18) Consider a floating-point system that uses 8-bit numbers: 3 bits for the exponent (with bias = 3) and 4 bits for the mantissa (excluding the implied leading 1 for normalized numbers). Illustrate the rounding method to represent the binary number 101.0011 in this floating-point system. (2)

19) Calculate the sum of two signed 8-bit binary numbers A= 11011001 and B=10110110 given in the two's complement representation. Is there an overflow in this addition? Justify your answer. (2)

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